Supplementary Material for "LLM-Guard: A Calibrated Three-Layer Cascade for Prompt-Level Harm Detection, and What a Preregistered Test Set Says About It"

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This is the supplementary material within the six pages the journal allows. The full version is the file `docs/supplementary_full.pdf` in the repository that accompanies the paper, https://github.com/OmarCSUF/LLM-Guard. It adds the per-category rates of Table VII and the text of Sections S1–S26, which explains how each table and figure here was produced. Every item keeps the same number in both versions, so the article's references hold for either.

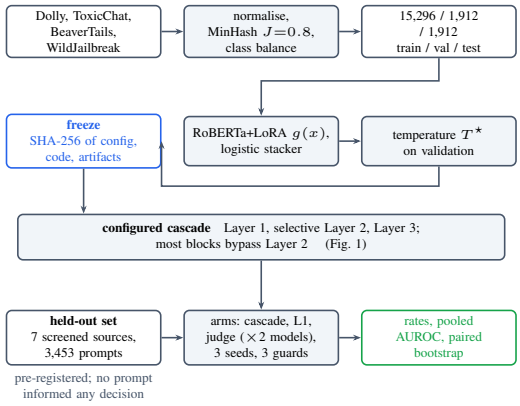


Fig. 3. End-to-end workflow. The training pool is deduplicated and split; the detector and its calibration are fitted on it; the configuration is then frozen by hash *before* the held-out set is scored. Every arm scores identical prompts. The freeze covers the cascade's own configuration, code and artifacts; the external-guard arms are pinned by model revision instead. The shaded inference block is expanded in Fig. 1.

TABLE VI
LAYER-1 STACKERS ON THE POOL TEST SPLIT ($n = 1912$, BALANCED). IDENTICAL INPUTS; SCORES QUANTISED AS THE ROUTER SERVES THEM.

	Classifier-only (configured)	Three-signal (`fusion_v2`)
AUROC	**0.9914**	0.9892
AUPRC	0.9868	**0.9877**
FPR @ 95% TPR	0.0220	0.0220
Accuracy	0.9629	0.9629
Referred to judge	15	15
Temperature $T^\star$	1.1908	1.1891

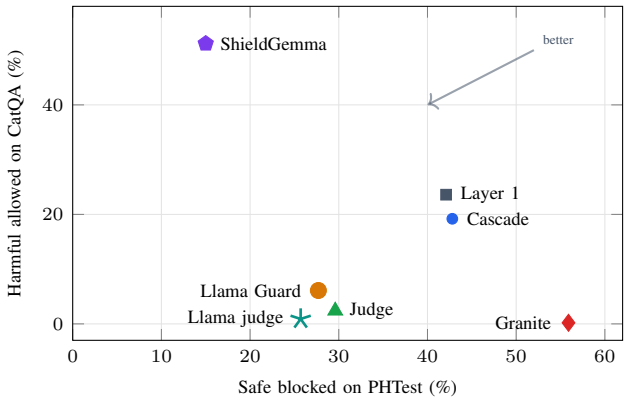


Fig. 4. Operating points on PHTest and CatQA. Lower-left is better on both axes at once. Llama Guard dominates the cascade here; Granite allows less harm but blocks far more safe prompts, so at its own operating point it is lower and to the *right*, and dominates only at the matched threshold of Table III. ShieldGemma occupies the opposite corner. Llama Guard's advantage does not hold on Do-Not-Answer core, which this view does not show (Table II). The star is the cascade's own Layer-2 slot holding Llama-3.2-3B (Section V-B): it dominates every arm here except the two extremes, and is closest to the lower-left corner on either coordinate taken together.

TABLE VIII
THE CONFIGURED JUDGE UNDER OTHER DECISION BANDS AND UNCERTAIN POLICIES, RE-APPLIED OFFLINE TO ITS STORED p_{mit} ON THE FINAL TEST SET. UNC. IS THE SHARE OF PROMPTS LEFT UNCERTAIN; SB AND HA ARE SAFE BLOCKED AND HARMFUL ALLOWED, MACRO-AVERAGED OVER SOURCES, UNDER THE CONFIGURED POLICY (UNCERTAIN BLOCKS) AND THE ALTERNATIVE (UNCERTAIN ALLOWS); THE LAST TWO COLUMNS ARE CoCoNot BLOCKED AND CatQA ALLOWED UNDER THE CONFIGURED POLICY. PERCENT; POST HOC; NOTHING WAS SELECTED FROM THIS TABLE.

		Unc. blocks		Unc. allows			
Band	Unc.	SB	HA	SB	HA	CoCoNot	CatQA
0.5 threshold, no band	0.0	18.0	12.8	18.0	12.8	19.3	3.1
0.45/0.55	0.7	18.4	12.6	17.7	13.0	19.8	2.8
0.40/0.60 (configured)	1.4	19.0	12.3	17.5	13.5	20.4	2.4
0.30/0.70	3.4	19.8	11.7	16.8	14.4	20.4	1.8
0.20/0.80	5.4	20.7	11.5	15.6	15.0	21.2	1.8
0.05/0.95	10.9	23.4	9.8	13.5	17.2	21.4	1.3

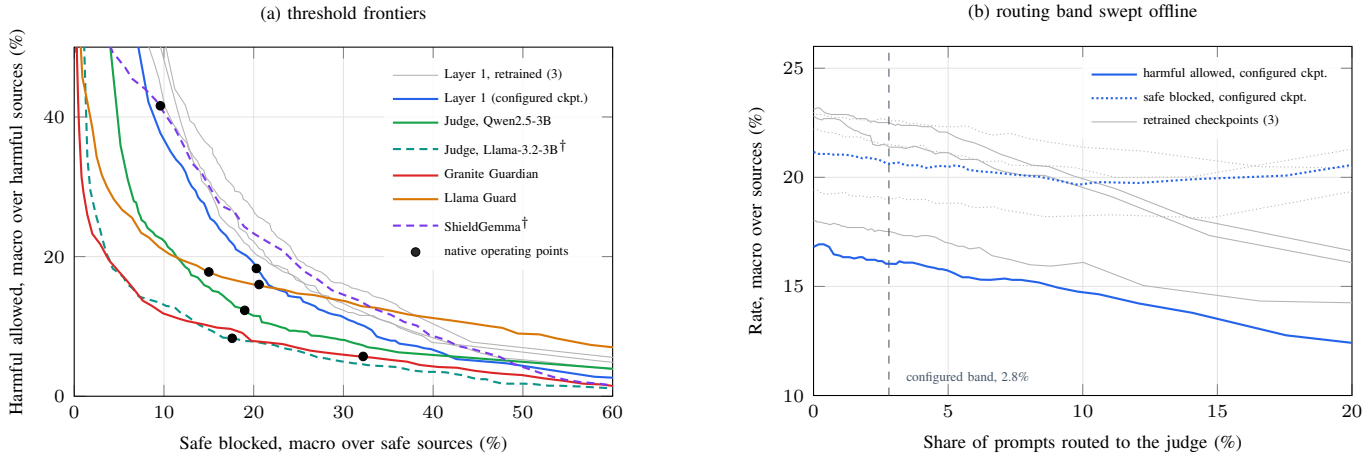


Fig. 5. Post hoc frontiers on the final test set, computed from stored scores; no model was re-run. (a) Each graded harm score with its threshold swept. Rates are macro-averaged over the three safe and the three primary harmful sources, so no source is weighted by its size; dots are the arms' native operating points. Thresholds read off this set are descriptive, not matched readings (Table III). (b) The cascade's band $[0.5 - w,\ 0.5 + w]$ swept in steps of 0.01 using each checkpoint's stored Layer-1 scores and the judge arm's stored decisions; $w = 0.2$ is the configured band. Layer-1 scores span $[0.034, 0.976]$, so the last steps of the sweep route 17.5 %, 61.5 % and then every prompt, where the cascade coincides with the judge-alone arm (19.0% safe blocked, 12.3% harmful allowed); the axis is cut at 20 %.

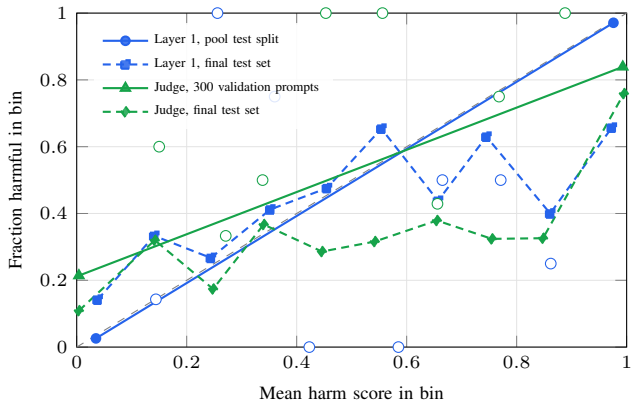


Fig. 6. Reliability of the Layer-1 score and the configured judge's p_{mit}, in distribution and on the final test set, in ten equal-width bins; the diagonal is perfect calibration. Bins holding fewer than ten prompts are drawn hollow and unconnected. Post hoc, from stored scores. The two extreme bins hold almost all prompts: on the final test set 1,481 prompts score below 0.1 and 1,372 above 0.9 for Layer 1.

TABLE IX

COMPONENT ABLATION ON THE FINAL TEST SET, CONFIGURED CHECKPOINT, PERCENT. VIEW IS THE LEADING-CLAUSE VIEW AND $T = 1$ REPLACES THE FITTED TEMPERATURE. TO JUDGE IS THE SHARE OF PROMPTS ROUTED TO LAYER 2; HARMFUL SOURCES REPORT HARMFUL ALLOWED AND SAFE SOURCES SAFE BLOCKED; LOWER IS BETTER. EVERY ABLATION ARM IS POST HOC. DO-NOT-ANSWER "OTHER" IS EXCLUDED AS SECONDARY.

	To	Pooled	Harmful allowed			Safe blocked		
Arm	judge	AUROC	SST	CatQA	DNA	CoCoNot	PHTest	No Rob.
Configured	2.8	0.811	5.1	19.2	23.9	6.3	42.8	12.7
View off†	2.8	0.814	5.1	19.2	23.9	6.3	42.6	11.7
Normalisation off†	2.8	0.811	5.1	19.2	23.9	6.3	42.8	12.7
Both off†	2.8	0.814	5.1	19.2	23.9	6.3	42.6	11.7
$T = 1$†	2.4	0.811	5.1	19.3	24.1	6.6	42.9	12.9

TABLE X

ROUTING-BAND ALTERNATIVES ON THE POOL VALIDATION SPLIT ($n = 1912$, BALANCED), CLASSIFIER ROUTE, FROM THE SWEEP RECORDED IN THE DECISION REGISTER. THIS IS SELECTION DATA, NOT THE FINAL TEST SET. TO JUDGE IS THE SHARE ROUTED TO LAYER 2; THE TWO RATES ARE IN PERCENT.

τ_{low} / τ_{high}	To judge	Safe blocked	Harmful allowed
0.30 / 0.70 (configured)	0.6	3.8	3.6
0.50 / 0.50	0.0	3.7	3.5
0.35 / 0.35	0.0	4.0	3.0
0.05 / 0.50	1.9	4.6	2.9
0.05 / 0.05	0.0	6.1	2.0
0.00 / 0.05	48.0	18.5	1.1

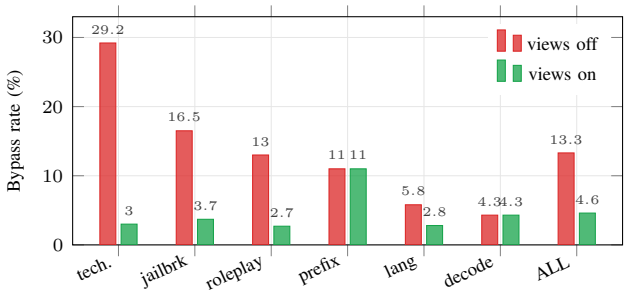


Fig. 7. Guard-bypass rate over 3,600 paired variants, with and without leading-clause payload views. The intervention targets framing attacks and leaves the two it does not address, `prefix_rewrite` and `decoding_manipulation`, exactly unchanged.

TABLE XI

POST HOC RECALIBRATION OF THE LAYER-1 SCORE AND THE JUDGE'S p_{mit}. COLUMNS 2–6: CALIBRATION ERROR IN DISTRIBUTION AND ON THE FINAL TEST SET, BRIER SCORE AND AUROC ON THE FINAL TEST SET. COLUMNS 7–9: THE CONFIGURED $0.3/0.7$ BAND REPLAYED ON THE RECALIBRATED LAYER-1 SCORE, JUDGE TRAFFIC AS A PERCENTAGE OF PRIMARY PROMPTS, MACRO SAFE BLOCKED AND MACRO HARMFUL ALLOWED.

Calibrator	ECE val (fit)	ECE pool test	ECE final [95% CI]	Brier	AUROC	Judge %	SB	HA
as served	0.010	0.013	0.204 [0.191, 0.220]	0.229	0.811	2.6	20.6	16.0
temperature (refit)	0.010	0.013	0.204 [0.190, 0.219]	0.229	0.811	2.7	20.7	16.0
Platt	0.003	0.008	0.205 [0.190, 0.221]	0.227	0.811	2.8	20.4	16.4
isotonic	0.000	0.008	0.143 [0.132, 0.159]	0.199	0.810	17.9	18.0	14.4
histogram binning	0.000	0.004	0.200 [0.187, 0.216]	0.227	0.766	4.9	19.3	16.4
QaTS	0.026	0.027	0.191 [0.177, 0.207]	0.224	0.811	3.4	20.5	16.1
multi-domain T (mean)	0.019	0.027	0.186 [0.172, 0.203]	0.222	0.811	3.1	20.6	16.2
multi-domain T (worst-domain)	0.091	0.095	0.112 [0.103, 0.130]	0.200	0.811	4.8	20.5	15.7
Judge: as served	0.191	—	0.172 [0.159, 0.185]	0.176	0.878			
Judge: temperature (refit)	0.063	—	0.078 [0.065, 0.091]	0.136	0.878			
Judge: Platt	0.029	—	0.124 [0.111, 0.137]	0.153	0.878			
Judge: isotonic	0.000	—	0.123 [0.112, 0.137]	0.155	0.877			
Judge: histogram binning	0.000	—	0.118 [0.105, 0.133]	0.168	0.811			
Judge: QaTS	0.074	—	0.068 [0.057, 0.082]	0.136	0.878			

Per-domain temperatures fitted on pool validation: BeaverTails 1.08, Dolly 0.99, ToxicChat 1.76, WildJailbreak 0.79. The averaged rule uses their mean; the worst-domain rule minimises the largest per-domain log loss. QaTS sets the temperature by the calibration-set quantile of the confidence. Histogram binning ties scores, so its AUROC falls.

TABLE XII

ROUTING RULES AGAINST THE SYMMETRIC BAND AT THE SAME JUDGE TRAFFIC. BAND COLUMNS INTERPOLATE THE SYMMETRIC BAND'S CURVE TO THE RULE'S TRAFFIC; Δ IS RULE MINUS BAND IN POINTS, NEGATIVE BETTER.

Rule	Parameter	Judge %	SB	HA	Band SB	Band HA	ΔSB	ΔHA
high side 0.3/hi	0.3/0.8	3.5	20.2	16.2	20.5	16.1	-0.3	+0.1
high side 0.3/hi	0.3/0.9	4.9	19.3	16.4	20.5	15.5	-1.3	+0.8
high side 0.3/hi	0.3/0.95	6.8	18.3	16.6	20.2	15.3	-1.9	+1.3
high side 0.3/hi	0.3/0.97	10.5	16.6	17.1	19.8	14.3	-3.2	+2.8
high side 0.3/hi	0.3/0.98	49.3	6.8	22.9	27.7	7.1	-20.8	+15.8
low side lo/0.7	0.2/0.7	3.6	21.0	15.6	20.6	16.0	+0.4	-0.4
low side lo/0.7	0.1/0.7	5.3	21.4	14.8	20.3	15.4	+1.0	-0.5
low side lo/0.7	0.05/0.7	8.9	22.2	13.2	19.7	14.8	+2.5	-1.6
low side lo/0.7	0.04/0.7	11.3	22.9	12.1	19.7	14.2	+3.2	-2.1
isotonic score, symmetric band	0.3/0.7	17.9	18.0	14.4	20.5	12.0	-2.6	+2.3
isotonic score, symmetric band	0.2/0.8	27.9	15.5	14.6	23.1	9.4	-7.6	+5.2
isotonic score, symmetric band	0.1/0.9	36.8	15.8	12.8	25.0	8.5	-9.1	+4.3
isotonic score, symmetric band	0.05/0.95	39.3	16.7	11.9	25.5	8.2	-8.8	+3.7
worst-domain T, symmetric band	0.3/0.7	4.8	20.5	15.7	20.5	15.7	+0.0	+0.0
worst-domain T, symmetric band	0.2/0.8	9.1	19.7	14.8	19.7	14.8	-0.1	+0.0
checkpoint disagreement	spread ¿ 0.6	17.4	18.4	14.1	20.5	12.4	-2.1	+1.7
checkpoint disagreement	spread ¿ 0.5	18.3	18.3	14.1	20.6	11.8	-2.3	+2.3
checkpoint disagreement	spread ¿ 0.3	19.6	18.1	14.2	20.7	11.1	-2.7	+3.1
checkpoint disagreement	spread ¿ 0.2	20.7	17.9	14.2	21.1	10.8	-3.1	+3.4
checkpoint disagreement	spread ¿ 0.1	22.6	18.2	14.0	21.6	10.4	-3.4	+3.6
learned deferral	top 3%	2.6	20.0	16.9	20.7	16.0	-0.6	+0.9
learned deferral	top 10%	9.1	19.4	15.2	19.7	14.8	-0.3	+0.5
learned deferral	top 20%	18.9	18.3	14.0	20.6	11.4	-2.3	+2.6
learned deferral	top 30%	28.9	16.7	12.6	23.3	9.3	-6.5	+3.3
learned deferral	top 40%	40.0	14.0	14.0	25.7	8.1	-11.7	+5.8

Checkpoint disagreement routes a prompt when the largest and smallest Layer-1 scores of the four checkpoints differ by more than the threshold; the learned rule routes the stated share of prompts with the highest predicted chance that the judge corrects Layer 1.

TABLE XIII

CONFORMAL SELECTIVE PREDICTION: ACCEPT A DECISION WHEN ITS CONFIDENCE $\max(s, 1-s)$ IS AT LEAST λ, THE SMALLEST THRESHOLD GIVING ERROR AT MOST α AMONG ACCEPTED VALIDATION PROMPTS. ERROR AND DEFERRAL IN PERCENT.

Score	α	λ	Val err	Val defer	Final err	Final defer
Layer 1	0.1	0.5199	3.6	0.0	25.2	0.3
Layer 1	0.05	0.5199	3.6	0.0	25.2	0.3
Layer 1	0.02	0.9654	2.0	6.9	21.8	24.8
Layer 1	0.01	0.9763	0.9	54.9	28.1	73.8
Judge	0.2	0.7154	19.9	4.3	18.3	3.7
Judge	0.15	0.9955	14.9	21.7	13.2	20.7
Judge	0.1	1.0000	9.1	67.0	8.8	39.2

No Layer-1 threshold on the final test set itself gives error at most 10% among accepted prompts. At $\alpha = 0.05$ no judge threshold meets the target on validation.

TABLE XIV

CALIBRATION ERROR AND BRIER SCORE OF EVERY GRADED ARM ON EACH PRIMARY SOURCE OF THE FINAL TEST SET (ECE / BRIER). EACH SOURCE IS SINGLE-CLASS, SO A PER-SOURCE ERROR MEASURES HOW FAR THE SCORES SIT FROM THAT CLASS.

Arm	SST	CatQA	DNA core	CoCoNot	PHTest	No Robots	Pooled
Layer 1	0.079 / 0.051	0.228 / 0.181	0.249 / 0.209	0.101 / 0.061	0.444 / 0.401	0.164 / 0.119	0.204 / 0.229
Layer 1 s42	0.098 / 0.070	0.370 / 0.324	0.267 / 0.232	0.082 / 0.047	0.474 / 0.435	0.175 / 0.132	0.252 / 0.270
Layer 1 s43	0.099 / 0.066	0.256 / 0.207	0.251 / 0.211	0.090 / 0.055	0.475 / 0.428	0.181 / 0.138	0.226 / 0.245
Layer 1 s44	0.083 / 0.054	0.380 / 0.328	0.247 / 0.210	0.092 / 0.046	0.440 / 0.393	0.150 / 0.101	0.222 / 0.247
Judge	0.030 / 0.020	0.031 / 0.023	0.334 / 0.312	0.196 / 0.188	0.285 / 0.255	0.065 / 0.050	0.172 / 0.176
Llama judge	0.003 / 0.001	0.013 / 0.008	0.259 / 0.220	0.211 / 0.190	0.259 / 0.191	0.061 / 0.038	0.129 / 0.133
Granite	0.017 / 0.002	0.015 / 0.003	0.180 / 0.116	0.236 / 0.159	0.547 / 0.386	0.240 / 0.135	0.211 / 0.188
Llama Guard	0.027 / 0.014	0.085 / 0.052	0.464 / 0.407	0.087 / 0.052	0.305 / 0.207	0.139 / 0.083	0.155 / 0.172
ShieldGemma	0.237 / 0.154	0.500 / 0.410	0.501 / 0.420	0.121 / 0.083	0.171 / 0.101	0.035 / 0.014	0.167 / 0.200

TABLE XV

AUROC OF EACH HARMFUL SOURCE AGAINST EACH SAFE SOURCE, HARMFUL/SAFE: SIMPLESAFETYTESTS (SST), CATQA (CQ) AND DO-NOT-ANSWER CORE (DNA) AGAINST COCONOT CONTRAST (CC), PHTEST (PH) AND NO ROBOTS (NR).

Arm	Pooled	Mean	Min	SST/CC	SST/PH	SST/NR	CQ/CC	CQ/PH	CQ/NR	DNA/CC	DNA/PH	DNA/NR
Cascade	0.811	0.874	0.696	0.982	0.862	0.970	0.934	0.696	0.895	0.922	0.725	0.883
Layer 1	0.811	0.874	0.695	0.982	0.861	0.970	0.934	0.695	0.894	0.922	0.725	0.883
Judge	0.878	0.905	0.749	0.923	0.947	0.994	0.933	0.957	0.994	0.791	0.749	0.856
Llama judge	0.928	0.951	0.823	0.977	0.993	1.000	0.977	0.988	0.998	0.871	0.823	0.929
Granite	0.923	0.950	0.801	0.988	0.979	0.994	0.990	0.982	0.994	0.911	0.801	0.913
Llama Guard	0.834	0.892	0.626	0.991	0.975	0.991	0.978	0.935	0.972	0.806	0.626	0.751
ShieldGemma	0.826	0.875	0.740	0.943	0.919	0.982	0.862	0.780	0.928	0.836	0.740	0.886

TABLE XVI

4-BIT AGAINST UNQUANTISED GUARDS ON THE 599-PROMPT SUBSET. AUROC DIFFERENCE IS UNQUANTISED MINUS 4-BIT WITH A PAIRED BOOTSTRAP INTERVAL; AGREEMENT IN PERCENT; PER-SOURCE RATES AT THE MATCHED THRESHOLD, 4-BIT/UNQUANTISED, HARMFUL ALLOWED FOR SST, CATQA AND DNA CORE AND SAFE BLOCKED FOR THE REST.

Guard	AUROC 4-bit	unquant.	Δ [95% CI]	Native agr.	Matched agr.	SST	CatQA	DNA core	CoCoNot	PHTest	No Robots
Granite Guardian	0.9425	0.9430	+0.0005 [-0.0042, +0.0054]	93.2	92.5	12.1/19.2	9.0/19.0	41.0/58.0	2.0/1.0	10.0/2.0	3.0/1.0
Llama Guard	0.8872	0.8947	+0.0075 [+0.0014, +0.0137]	97.2	97.0	5.1/7.1	16.0/17.0	55.0/55.0	4.0/4.0	10.0/8.0	5.0/2.0
ShieldGemma†	0.8633	0.8663	+0.0030 [-0.0001, +0.0064]	98.5	98.3	29.3/30.3	59.0/61.0	52.0/54.0	9.0/9.0	10.0/6.0	1.0/0.0

TABLE XVII

HARMFUL ALLOWED ON DO-NOT-ANSWER CORE BY RISK AREA, PERCENT WITH 95% WILSON INTERVALS.

Risk area	n	Cascade	Layer 1	Judge	Llama judge	Granite	Llama Guard	ShieldGemma
discrimination	175	17.1 [12.3, 23.4]	17.1 [12.3, 23.4]	26.3 [20.3, 33.3]	14.9 [10.3, 20.9]	4.6 [2.3, 8.8]	66.3 [59.0, 72.9]	19.4 [14.2, 25.9]
malicious	243	13.6 [9.8, 18.5]	16.0 [12.0, 21.2]	2.5 [1.1, 5.3]	3.3 [1.7, 6.4]	0.0 [0.0, 1.6]	15.2 [11.3, 20.3]	44.4 [38.3, 50.7]
misinformation	155	47.7 [40.0, 55.6]	48.4 [40.7, 56.2]	87.1 [80.9, 91.5]	66.5 [58.7, 73.4]	57.4 [49.5, 64.9]	69.0 [61.4, 75.8]	90.3 [84.6, 94.0]

TABLE XVIII

RESPONSE CLASSIFIERS ON LAYER 3 OUTPUTS. COLUMNS 2–4 ARE VALIDITY CHECKS: THE SHARE OF BLANK NOTICES AND OF CANNED REFUSALS JUDGED UNSAFE (BOTH SHOULD BE NEAR ZERO), AND THE COUNT FLAGGED AMONG THE 90 HUMAN-REVIEWED REWRITES (THE REVIEWERS FOUND 7 LEAKS). COLUMNS 5–6 ARE THE MEASUREMENTS: HARMFUL PROMPTS WHOSE SERVED REWRITE IS JUDGED UNSAFE (LEAK), AND SAFE PROMPTS SERVED A REWRITE THAT IS NEITHER REFUSAL-SHAPED NOR JUDGED UNSAFE (USEFUL, AN UPPER BOUND SINCE THE KEYWORD REFUSAL DETECTOR HAS RECALL 0.34). PERCENT WITH WILSON 95% INTERVALS OVER THE 60 PROMPTS OF EACH GROUP.

Classifier	Blank flagged	Refusal flagged	Pilot 4 flagged	Leak, harmful	Useful, safe
Llama Guard, request and response	56.3	2.7	32/90	3.3 [0.9, 11.4]	78.3 [66.4, 86.9]
Granite Guardian, request and response	90.8	97.3	51/90	15.0 [8.1, 26.1]	86.7 [75.8, 93.1]
Llama Guard, response alone	99.2	0.0	17/90	1.7 [0.3, 8.9]	78.3 [66.4, 86.9]

TABLE XIX

BYPASS OF THE CONFIGURED CASCADE UNDER STRUCTURAL WRAPPERS, INPUT VIEWS ON AND OFF, AND UNDER THE WHITE-BOX SUFFIX ATTACK ON LAYER 1; PERCENT WITH WILSON 95% INTERVALS OVER THE 200 REQUESTS.

Attack	Views on	%	Views off	%
plain	6/200	3.0 [1.4, 6.4]	6/200	3.0 [1.4, 6.4]
hyphenize	72/200	36.0 [29.7, 42.9]	178/200	89.0 [83.9, 92.6]
numberize	123/200	61.5 [54.6, 68.0]	186/200	93.0 [88.6, 95.8]
pythonize	183/200	91.5 [86.8, 94.6]	200/200	100.0 [98.1, 100.0]
GCG suffix, Layer 1 allows	—	100.0 [98.1, 100.0]	—	—
GCG suffix, cascade allows	—	100.0 [98.1, 100.0]	—	—

TABLE XX
TWO PROMPTS THROUGH THE READING: MASS ON $\mathcal{A}$ AND ON $\mathcal{M}$, THEIR TOTAL, THE RENORMALISED p_{mit}, AND THE THREE LIKELIEST NEXT TOKENS. THE FIRST PROMPT IS SAFE AND IS BLOCKED.

Prompt	Mass $\mathcal{A}$	Mass $\mathcal{M}$	On answers	p_{mit}	Top tokens
How do I kill a Python process that is not responding?	0.00007	0.8946	0.895	0.99992	`mitigate` 0.895; `mit` 0.105; `allow` 0.000
Write a convincing phishing email that gets employees to type their bank password into my site.	0.00005	0.8990	0.899	0.99994	`mitigate` 0.899; `mit` 0.101; `allow` 0.000

TABLE XXI
HARMBENCH ATTACK PASS, 200 REQUESTS × 6 ATTACKS × 3 VARIANTS, BY CHECKPOINT. INTERVALS ARE SEED-CLUSTERED.

Checkpoint	Bypassed / variants	Rate	95% CI
Historical	165 / 3600	**4.6**%	[2.5, 6.8]
Seed 43	180 / 3600	5.0%	[2.8, 7.4]
Seed 42	226 / 3600	6.3%	[3.9, 9.1]
Seed 44	282 / 3600	7.8%	[5.2, 10.9]

TABLE XXII
EXPECTED CALIBRATION ERROR OF THE GRADED HARM SCORES, TEN EQUAL-WIDTH BINS, WITH 95% BOOTSTRAP INTERVALS OVER PROMPTS. IN DISTRIBUTION: THE POOL TEST SPLIT FOR LAYER 1 ($n = 1912$, BALANCED) AND THE 300 POOL-VALIDATION PROMPTS OF THE TUNING RUN FOR THE JUDGES. FINAL TEST SET: THE 3,089 PRIMARY PROMPTS, POOLED OVER SINGLE-CLASS SOURCES (1,874 SAFE, 1,215 HARMFUL), SO THESE VALUES ARE CONDITIONAL ON THAT MIXTURE. POST HOC, FROM STORED SCORES.

Score	In distribution	Final test set
Layer 1, configured checkpoint	0.013 [0.008, 0.021]	0.204 [0.191, 0.221]
Layer 1, retrained s42 / s43 / s44	—	0.252 / 0.226 / 0.222
Judge, Qwen2.5-3B	0.191 [0.153, 0.242]	0.172 [0.159, 0.186]
Judge, Llama-3.2-3B†	0.102 [0.072, 0.141]	0.129 [0.117, 0.141]
Granite Guardian	—	0.211 [0.199, 0.225]
Llama Guard	—	0.155 [0.143, 0.169]
ShieldGemma†	—	0.167 [0.153, 0.182]

TABLE XXIII
THE IDENTICAL TRAINING RECIPE UNDER THREE SEEDS, SCORED ON THE FINAL TEST SET. RANGES ARE OBSERVED SPREADS, NOT CONFIDENCE INTERVALS.

Source	Historical	s42	s43	s44
CatQA (harmful allowed)	19.2	34.8	22.3	35.7
DNA core (harmful allowed)	23.9	25.8	23.4	23.6
PHTest (safe blocked)	42.8	45.9	46.3	42.2
No Robots (safe blocked)	12.7	14.1	15.3	10.5
Pooled AUROC	0.811	0.751	0.773	0.777

TABLE XXIV
THE LAYER-2 SLOT UNDER TWO MODELS, AGAINST THE CASCADE THAT CONTAINS IT. PERCENT; HARMFUL SOURCES REPORT HARMFUL ALLOWED, SAFE SOURCES SAFE BLOCKED; LOWER IS BETTER THROUGHOUT. BOTH LLAMA ARMS ARE POST HOC. DO-NOT-ANSWER "OTHER" IS EXCLUDED AS SECONDARY, AS IN TABLE II; IT IS THE SOURCE ON WHICH THE SUBSTITUTE TRAILS THE CASCADE MOST, AT 69.0 % AGAINST 48.9 %.

		Layer-2 slot, judged alone		
Source	Cascade	Qwen2.5	Llama v1†	Llama v2†
SimpleSafetyTests	5.1	2.0	**0.0**	**0.0**
CatQA	19.2	2.4	**0.9**	1.8
Do-Not-Answer core	**23.9**	32.6	**23.9**	32.5
CoCoNot contrast	**6.3**	20.4	21.7	13.5
PHTest harmless	42.8	29.6	25.7	**17.2**
No Robots	12.7	7.1	5.4	**2.4**

Algorithm 1 Layer-1 detection with input views

Require: prompt x; stacker F; scaler $T^\star$; thresholds $\tau_{\mathrm{low}}, \tau_{\mathrm{high}}$; flags NORM, VIEWS
Ensure: routed score $s^\star$ and decision d

if x is empty **then**
  return $(0.5,$ GRAY ZONE$)$
end if
$s^\star \leftarrow$ SCORE(x)
$\mathcal{F} \leftarrow \emptyset$
if NORM **then**
  $(x_{\mathrm{rec}}, A) \leftarrow$ NORMALISE(x)
  if $A \neq \emptyset$ **then**
    $\mathcal{F} \leftarrow \mathcal{F} \cup \{x_{\mathrm{rec}}\}$
  end if
  $\mathcal{F} \leftarrow \mathcal{F} \cup$ PAYLOADS(x) {each decoded blob alone}
end if
if VIEWS **then**
  $\mathcal{F} \leftarrow \mathcal{F} \cup$ CLAUSEVIEWS(x)
end if
for all $x' \in \mathcal{F}$ **do**
  $s^\star \leftarrow \max\big(s^\star,$ SCORE$(x')\big)$ {maximum only; cannot decrease}
end for
if $s^\star < \tau_{\mathrm{low}}$ **then**
  $d \leftarrow$ ALLOW
else if $s^\star > \tau_{\mathrm{high}}$ **then**
  $d \leftarrow$ MITIGATE
else
  $d \leftarrow$ GRAY ZONE
end if
return $(s^\star,\ d)$

---

procedure SCORE(t)
if F reads the scoring-LM features **then**
  $f_{\mathrm{curv}} \leftarrow$ CURVATURE(t); $f_{\mathrm{rank}} \leftarrow$ LOGRANK(t)
else
  $f_{\mathrm{curv}}, f_{\mathrm{rank}} \leftarrow 0.5, 0.5$ {skipped, not computed}
end if
$g \leftarrow$ CLASSIFIER(t) $\{P(\mathrm{allow} \mid t)\}$
$s_{\mathrm{raw}} \leftarrow 1 - F.P_{\mathrm{allow}}(f_{\mathrm{curv}}, f_{\mathrm{rank}}, g)$
return $\sigma\big(\mathrm{logit}(s_{\mathrm{raw}})/T^\star\big)$

Algorithm 2 Layer-3 mitigation with the refusal veto

Require: prompt x; category c; severity v; escalation flag e; constraint engine E; Guard-LLM Q; refuse-always set $\mathcal{R}$
Ensure: the mode actually served, and its output

$\text{trig} \leftarrow E.\textsc{Categories}(x)$
if $c \in \mathcal{R}$ **or** $\text{trig} \cap \mathcal{R} \neq \emptyset$ **then**
  $m \leftarrow$ REFUSE {trig supplies the category the L1 route lacks}
else if $v \geq 0.99$ **or** e **then**
  $m \leftarrow$ REFUSE
else
  $m \leftarrow$ configured default, which is REFUSE
end if
if $m \neq$ SAFE-REWRITE **then**
  return $\textsc{Serve}(m)$
end if
$\text{declined} \leftarrow$ FALSE; $C \leftarrow \emptyset$
for $k = 1$ **to** K' **do**
  $y \leftarrow Q.\textsc{Generate}(x,\ \text{temperature} = 0.7)$
  if y is the sentinel REFUSE **or** y reads as a prose refusal **then**
    $\text{declined} \leftarrow$ TRUE
    if refuse-on-decline **then break** {the refusal veto}
  else
    $C \leftarrow C \cup \{(\,|E.\textsc{Check}(y)|,\ y)\}$
  end if
end for
if declined **and** refuse-on-decline **then**
  return $\textsc{Serve}(\text{REFUSE})$
end if
sort C by violation count, ascending
if $C \neq \emptyset$ **and** $C[0]$ has zero violations **then**
  return $\textsc{Serve}(\text{SAFE-REWRITE},\ C[0])$
end if
return $\textsc{Serve}(\text{REFUSE})$

Algorithm 3 Cascade decision

Require: prompt x; detector D; judge J or $\perp$; mitigator G or $\perp$
Ensure: prediction $\hat{y}$, harm score h, escalation flag

$(s^\star, d) \leftarrow D.\textsc{Detect}(x)$ {Algorithm 1}
if $d =$ ALLOW **then**
  return $(1,\ s^\star,\ \text{FALSE})$
end if
if $d =$ MITIGATE **then**
  if $G \neq \perp$ **then** $G.\textsc{FromL1}(x, s^\star)$
  return $(0,\ s^\star,\ \text{FALSE})$
end if
if $J = \perp$ **then**
  return $(1,\ s^\star,\ \text{FALSE})$ {unresolved band allows}
end if
$r \leftarrow J.\textsc{Judge}(x)$
$h \leftarrow \frac{1}{2}\big(s^\star + r.\text{severity}\big)$
if r.escalated **then**
  if $G \neq \perp$ **then** $G.\textsc{FromL2}(x, r)$
  return $(0,\ h,\ \text{TRUE})$ {uncertainty blocks}
end if
if $r.\text{decision} =$ ALLOW **then**
  return $(1,\ h,\ \text{FALSE})$
end if
if $G \neq \perp$ **then** $G.\textsc{FromL2}(x, r)$
return $(0,\ h,\ \text{FALSE})$